THE SCHEME THEORETIC SUPPORT OF A FLAT PULLBACK IS THE PREIMAGE OF THE SUPPORT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof names both characterizations that make the reduction legitimate; the flatness hypothesis is used exactly to transfer the scheme statement back.

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- The characterization of the scheme theoretic support of a finite type module. Lemma lemma-scheme-theoretic-support, used at proof lines 6254-6255. Identifies the two sides of the claimed equality.
- Flatness characterized through etale coverings. Lemma lemma-flat-local, used at proof lines 6256-6257. Reduces flatness of f to the scheme case.
- The scheme case for the support of a flat pullback. Morphisms, Lemma morphisms-lemma-flat-pullback-support, used at proof lines 6258-6259. Is the input that closes the argument.

Source reference. Original lemma 03MM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4754–4754. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 07U0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9955–9955. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 07T9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9929–9929. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $f : Y \to X$ be a morphism of algebraic spaces over S. Let $\mathcal{F}$ be a finite type quasi-coherent $\mathcal{O}_X$-module with scheme theoretic support $Z \subset X$. If f is flat, then $f^{-1}(Z)$ is the scheme theoretic support of $f^*\mathcal{F}$.*

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Using the characterization of the scheme theoretic support as given in Lemma 07U0 and using the characterization of flat morphisms in terms of étale coverings in Lemma 03MM we reduce to the case of schemes which is Morphisms, Lemma 07T9. □